

Performance Testing

Date	08 July 2026
Team ID	XXXXXX
Project Name	Smart Lender - AI Powered Loan Eligibility Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the Smart Lender system behaves under expected load conditions, and how accurately the deployed model performs on unseen test data.

Step 1: Testing Overview

Testing Tool Used	Postman / manual browser testing
Type of Testing	Load Testing & Model Accuracy Testing
Target Module / Route	/predict (Flask)
Test Environment	Local system (Flask development server)
Test Date	08-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Open Home Page	1	30 sec	Page loads successfully with the loan application form
2	Submit valid loan application	5	60 sec	Prediction generated correctly on the result page
3	Multiple concurrent requests	10	60 sec	System remains responsive under simultaneous predictions
4	Submit invalid / missing input	5	30 sec	Application returns a clear error instead of crashing

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status
1	Response Time (Avg)	< 2 seconds	0.4 sec	Pass
2	Response Time (Max)	< 5 seconds	1.1 sec	Pass
3	Throughput (Req/sec)	20 req/sec	24 req/sec	Pass
4	Error Rate	< 1%	0%	Pass
5	CPU Utilization	< 80%	38%	Pass
6	Memory Utilization	< 80%	47%	Pass

Model Accuracy Testing (Test Dataset)

The trained KNN model was evaluated on the held-out test split (123 records, 20% of the dataset). The results below were produced directly from models/loan_model.pkl against datasets/X_test.csv and datasets/y_test.csv.

Metric	Value
Accuracy	84.55%
Precision (Approved)	0.84
Recall (Approved)	0.96
F1-score (Approved)	0.90

